Replay-Aware TSP Aggregate Report

Overview

- Run count: 20.
- Condition count: 4.
- Best mean transfer gap: tsplib_random_replay.

Condition Comparison

Condition	Mean Transfer Gap	Mean TSPLIB Gap	Mean Synthetic Gap	Mean Accepted Novelty	Mean Complexity	Mean Adaptation Efficiency
tsplib_failure_replay	0.130918	0.18132	0.042713	0.806087	0.631	-8.5e-05
tsplib_failure_replay_compression	0.124895	0.177578	0.032698	0.286629	0.67	0.015348
tsplib_no_replay	0.132474	0.183325	0.043485	0.742512	0.63	-0.01061
tsplib_random_replay	0.104243	0.152211	0.0203	0.736465	0.7	0.025089

Bootstrap Confidence Intervals

tsplib_failure_replay

- Transfer gap 95% CI: 0.111752 to 0.148278.
- TSPLIB gap 95% CI: 0.158054 to 0.201666.
- Synthetic gap 95% CI: 0.029212 to 0.055179.
- Mean accepted novelty 95% CI: 0.708909 to 0.868262.

tsplib_failure_replay_compression

- Transfer gap 95% CI: 0.106918 to 0.141737.
- TSPLIB gap 95% CI: 0.155253 to 0.19875.
- Synthetic gap 95% CI: 0.019743 to 0.045373.
- Mean accepted novelty 95% CI: 0.13723 to 0.445475.

tsplib_no_replay

- Transfer gap 95% CI: 0.113762 to 0.14929.
- TSPLIB gap 95% CI: 0.160588 to 0.203404.
- Synthetic gap 95% CI: 0.030349 to 0.055951.
- Mean accepted novelty 95% CI: 0.617775 to 0.839715.

tsplib_random_replay

- Transfer gap 95% CI: 0.08686 to 0.122292.
- TSPLIB gap 95% CI: 0.130868 to 0.173495.

- Synthetic gap 95% CI: 0.007626 to 0.033284.
- Mean accepted novelty 95% CI: 0.643003 to 0.797918.

Paired Comparisons

Negative delta favors the candidate for transfer gap, benchmark gap, synthetic gap, novelty, and complexity. Positive delta favors the candidate for adaptation efficiency.

Comparison	Paired Runs	Transfer Delta	Transfer 95% CI	TSPLIB Delta	TSPLIB 95% CI	Novelty Delta	Complexity Delta	Adaptation Delta
tsplib_random_replay vs tsplib_no_replay	20	-0.028231	-0.052034 to -0.002934	-0.031114	-0.060232 to -0.000821	-0.006048	0.07	0.035699
tsplib_failure_replay vs tsplib_no_replay	20	-0.001556	-0.023279 to 0.021602	-0.002004	-0.027988 to 0.024793	0.063575	0.001	0.010525
tsplib_failure_replay vs tsplib_random_replay	20	0.026674	-0.005591 to 0.056743	0.02911	-0.010083 to 0.064428	0.069622	-0.069	-0.025174
tsplib_failure_replay_compression vs tsplib_failure_replay	20	-0.006023	-0.030618 to 0.018585	-0.003742	-0.035531 to 0.027774	-0.519458	0.039	0.015433
tsplib_failure_replay_compression vs tsplib_no_replay	20	-0.00758	-0.031668 to 0.016734	-0.005747	-0.035992 to 0.024612	-0.455883	0.04	0.025957

Paired Notes

- tsplib_random_replay vs tsplib_no_replay used 20 paired runs.
- Mean transfer-gap delta: -0.028231 with 95% CI -0.052034 to -0.002934; candidate better on 55.0% of paired runs.
- Mean TSPLIB-gap delta: -0.031114 with 95% CI -0.060232 to -0.000821.
- tsplib_failure_replay vs tsplib_no_replay used 20 paired runs.
- Mean transfer-gap delta: -0.001556 with 95% CI -0.023279 to 0.021602; candidate better on 35.0% of paired runs.
- Mean TSPLIB-gap delta: -0.002004 with 95% CI -0.027988 to 0.024793.
- tsplib_failure_replay vs tsplib_random_replay used 20 paired runs.
- Mean transfer-gap delta: 0.026674 with 95% CI -0.005591 to 0.056743; candidate better on 20.0% of paired runs.
- Mean TSPLIB-gap delta: 0.02911 with 95% CI -0.010083 to 0.064428.
- tsplib_failure_replay_compression vs tsplib_failure_replay used 20 paired runs.
- Mean transfer-gap delta: -0.006023 with 95% CI -0.030618 to 0.018585; candidate better on 35.0% of paired runs.
- Mean TSPLIB-gap delta: -0.003742 with 95% CI -0.035531 to 0.027774.
- tsplib_failure_replay_compression vs tsplib_no_replay used 20 paired runs.
- Mean transfer-gap delta: -0.00758 with 95% CI -0.031668 to 0.016734; candidate better on 30.0% of paired runs.
- Mean TSPLIB-gap delta: -0.005747 with 95% CI -0.035992 to 0.024612.